EXPERIMENT1: GETTING STARTED WITH POWER BI DESKTOP

1.Aim

To understand the **basic interface of Microsoft Power BI Desktop and create a simple report using data visualization tools.

2. Objectives

After completing this experiment, students will be able to:

* Understand the Power BI interface

* Import data into Power BI

* Create visualizations like charts and tables

* Save reports in Power BI format

Power BI is a *business intelligence tool used to convert raw data into meaningful insights using dashboards and reports.

3. Software Requirements

* Windows 10 / Windows 11

* Internet Connection

* Microsoft Power BI Desktop

* Sample Excel Dataset

4. Theory

Power BI is a *data analysis and visualization tool developed by Microsoft*. It allows users to connect to different data sources, clean the data, and create interactive reports and dashboards.

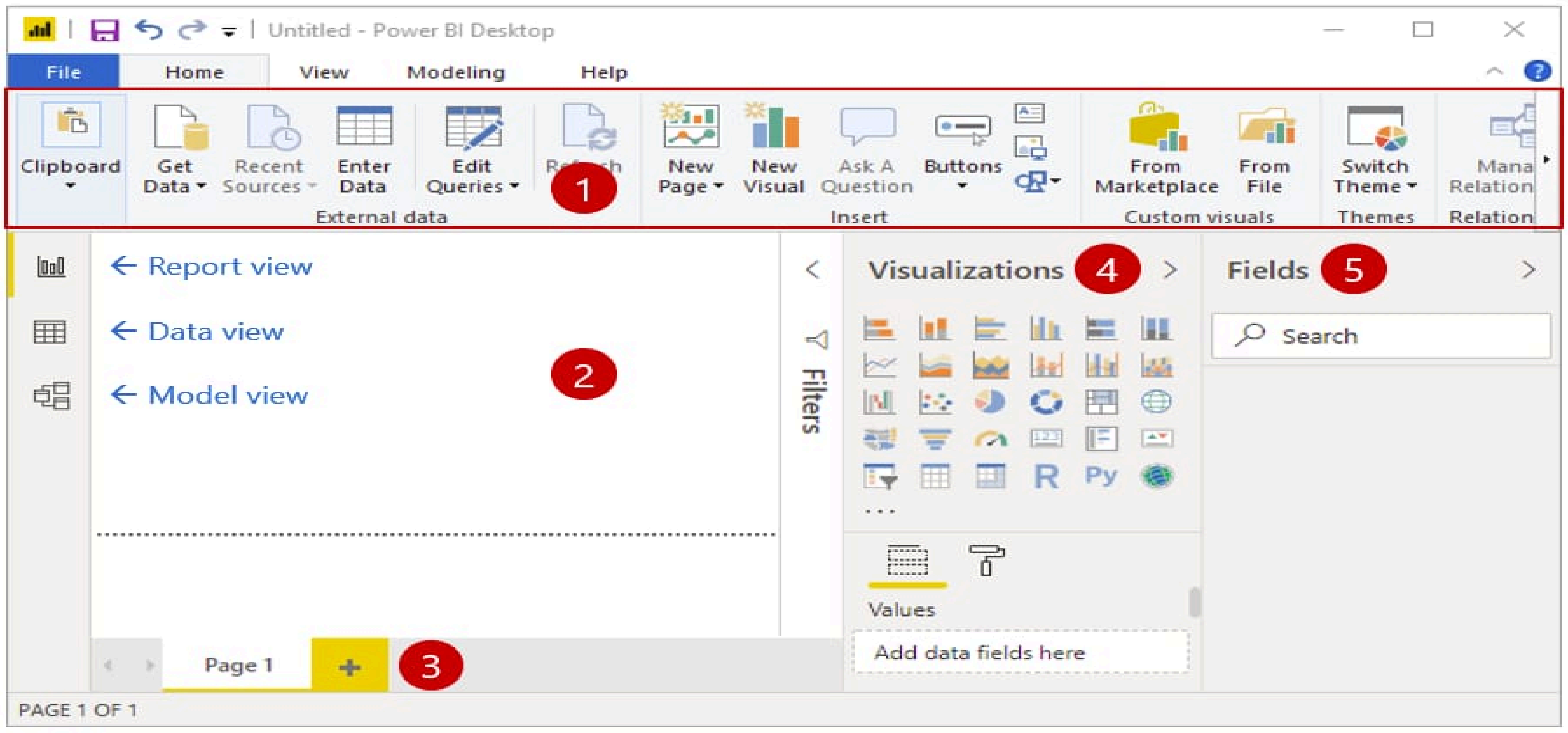
Main components of Power BI:

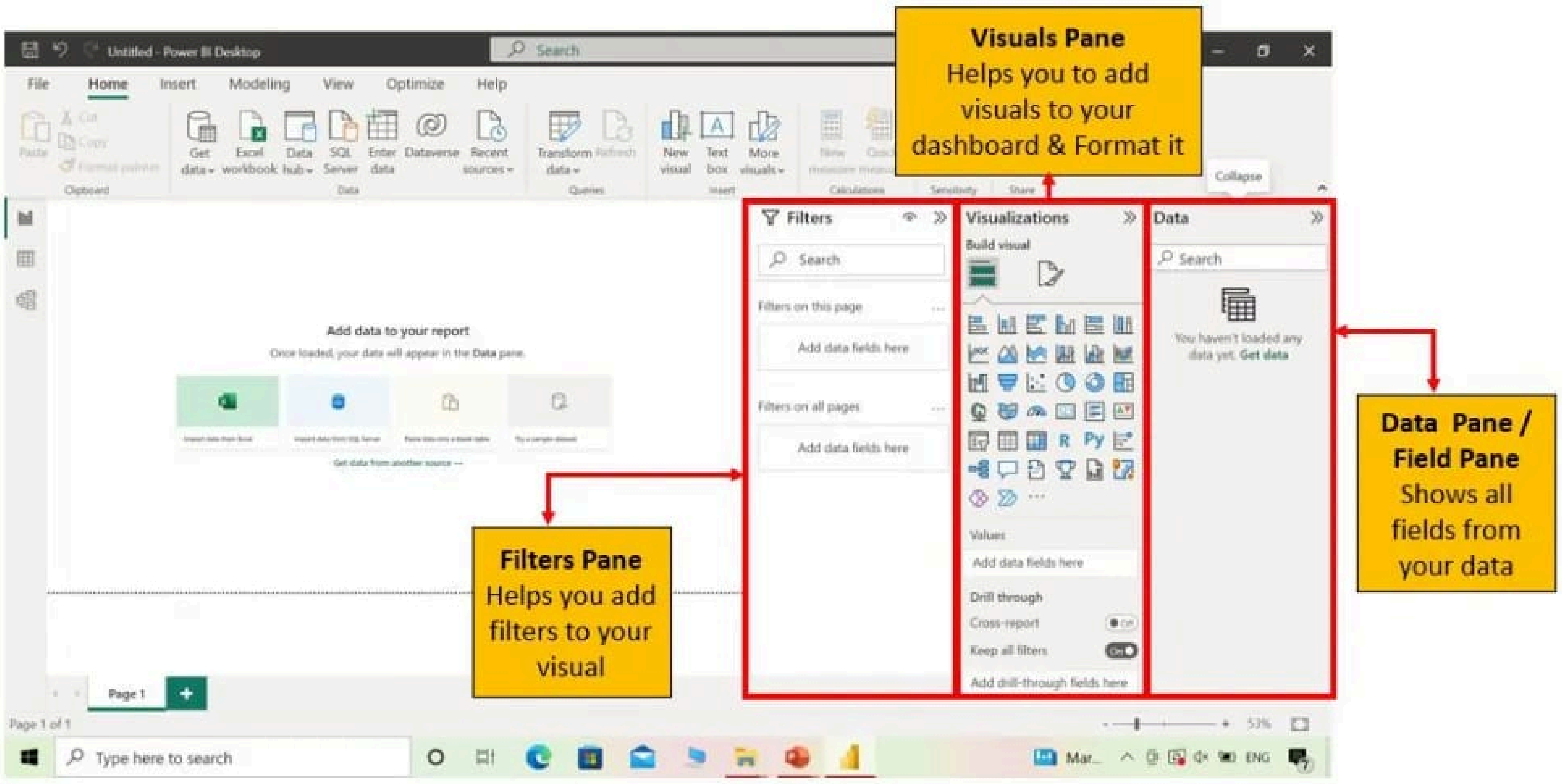
1. Power BI Desktop – Report creation

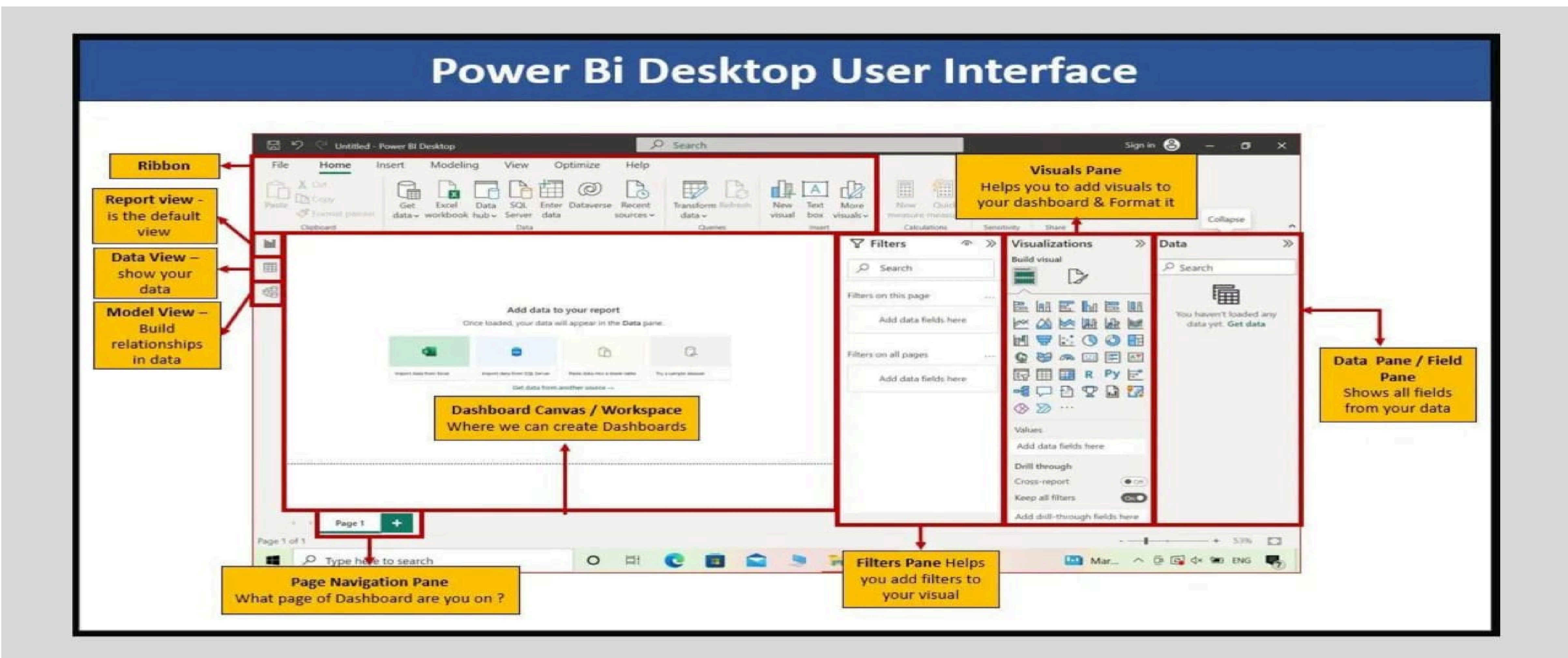
2. Power BI Service – Online report sharing

3. Power BI Mobile – View reports on mobile devices

5. Power BI Interface







Important parts of the interface:

1. *Ribbon* – Contains commands like Get Data, Transform Data

2. *Report Canvas* – Area where visualizations are created

3. *Visualizations Pane* – Contains charts and graphs

4. *Fields Pane* – Displays imported dataset fields

5. *Pages Tab* – Used to create multiple report pages

6. Procedure

Step 1: Install Power BI

1. Open browser

2. Search *Power BI Desktop Download*

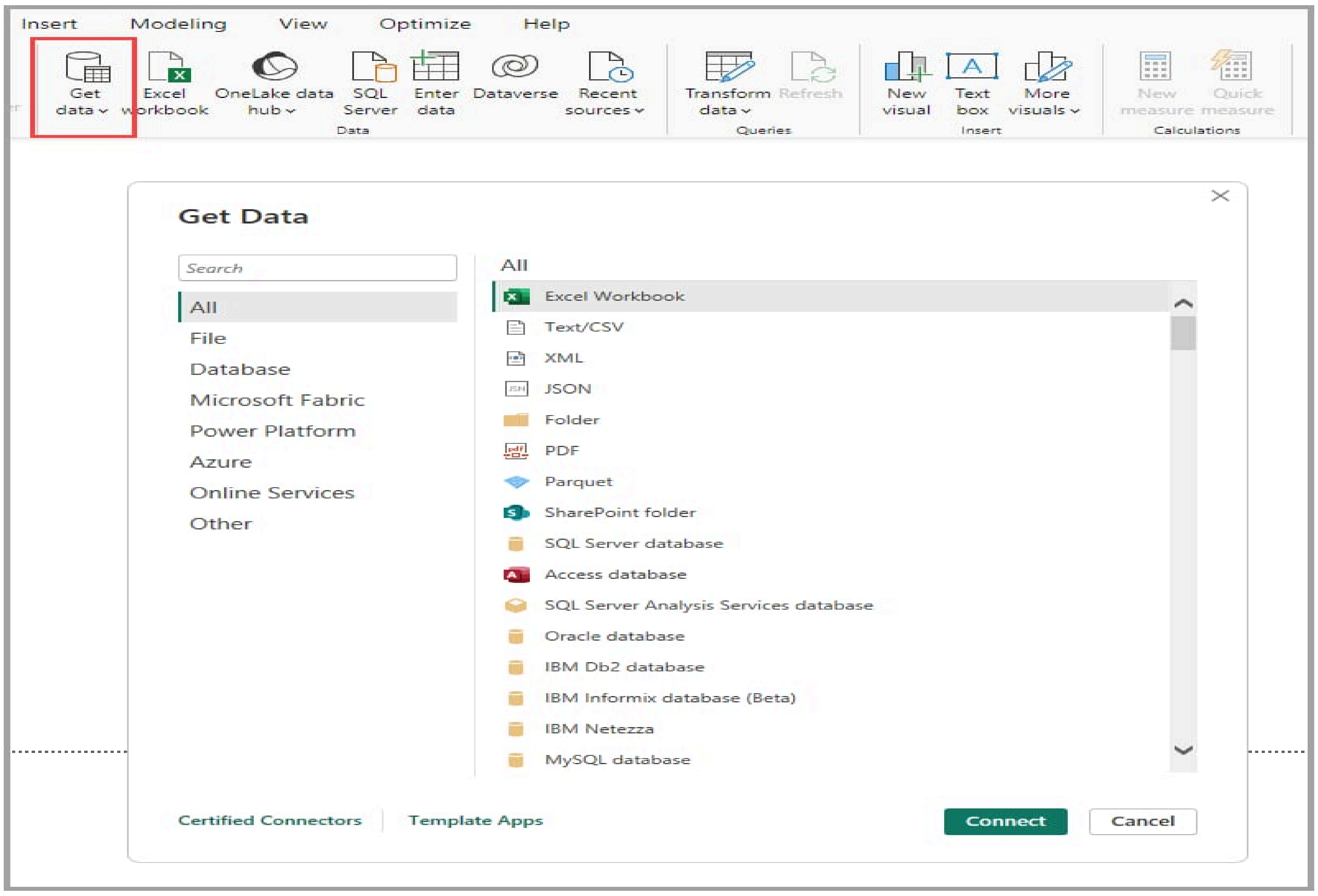
3. Download and install the softwar

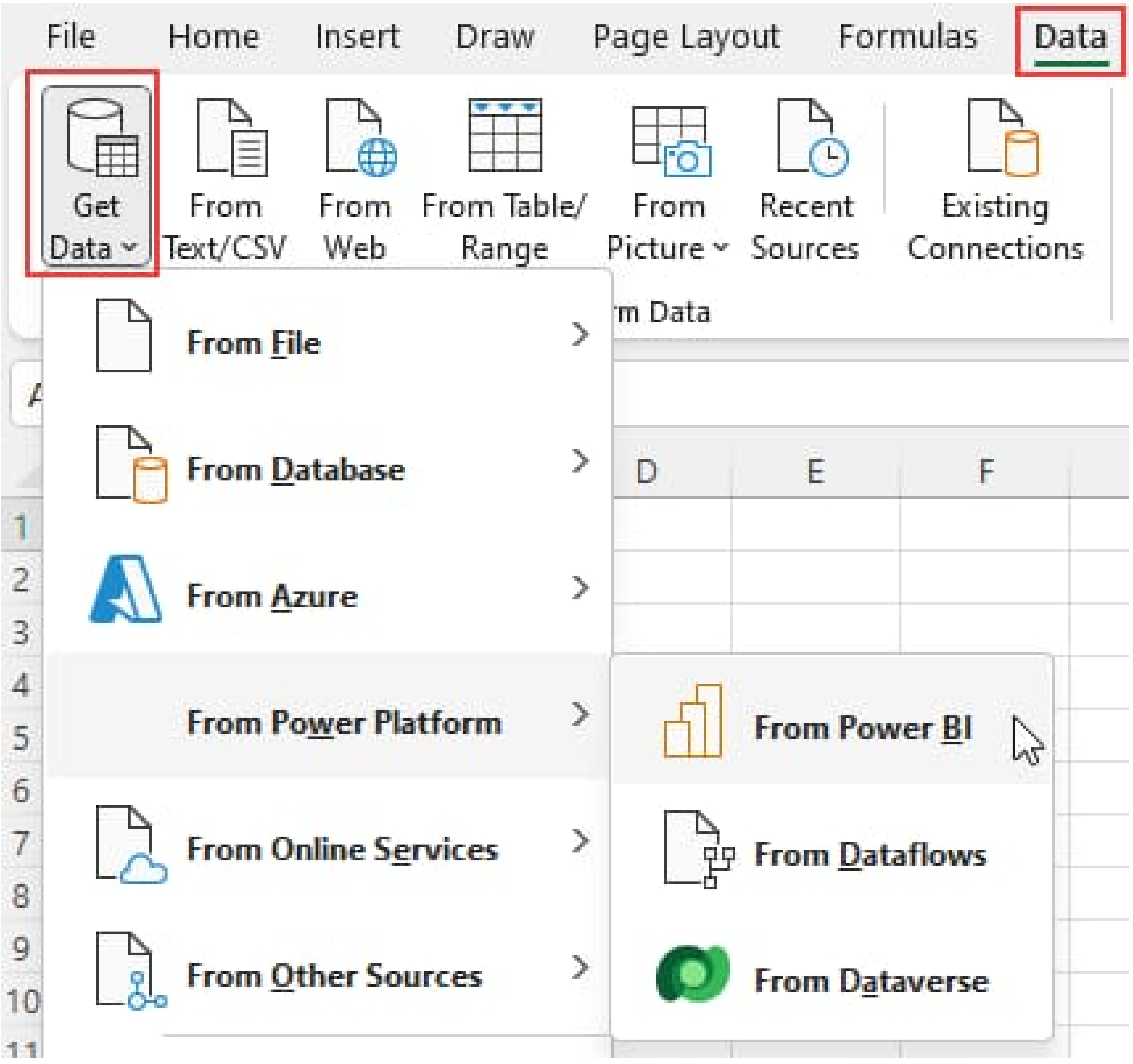
Step 2: Open Power BI Desktop

1. Launch Power BI Desktop

2. Click *Home Tab*

Step 3: Import Data





Steps:

1. Click *Get Data*

2. Select *Excel / CSV*

3. Choose dataset file

4. Click *Load*

The dataset will appear in the *Fields Panel*.

Step 4: Transform Data

If the data needs cleaning:

1. Click *Transform Data*

2. Power Query Editor will open

3. Edit or clean data

4. Click *Close & Apply*

Step 5: Create Visualizations

1. Select a visualization from the *Visualizations Pane*

2. Drag fields into the chart area

3. Add Axis, Values, and Legend

Examples:

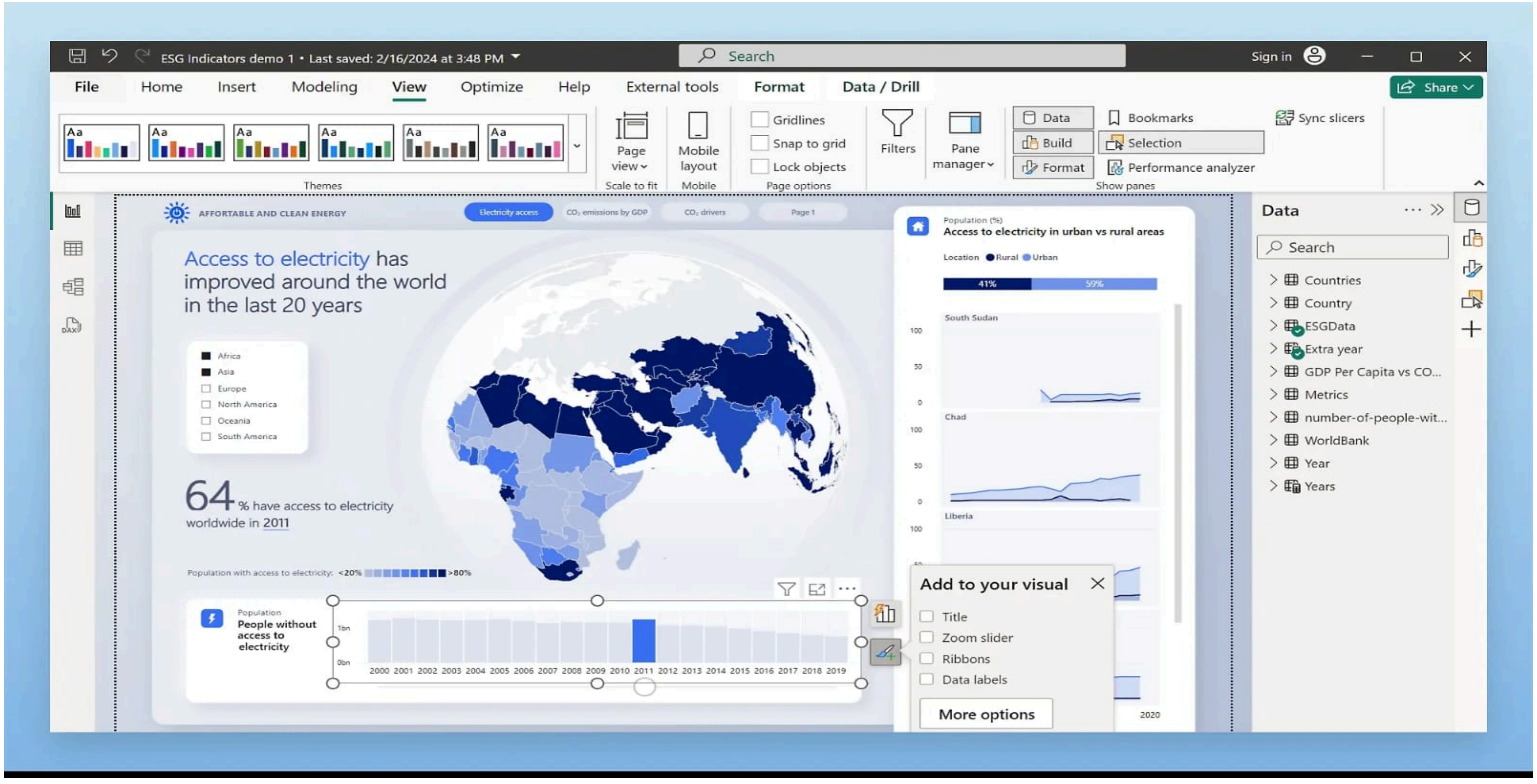
* Bar Chart

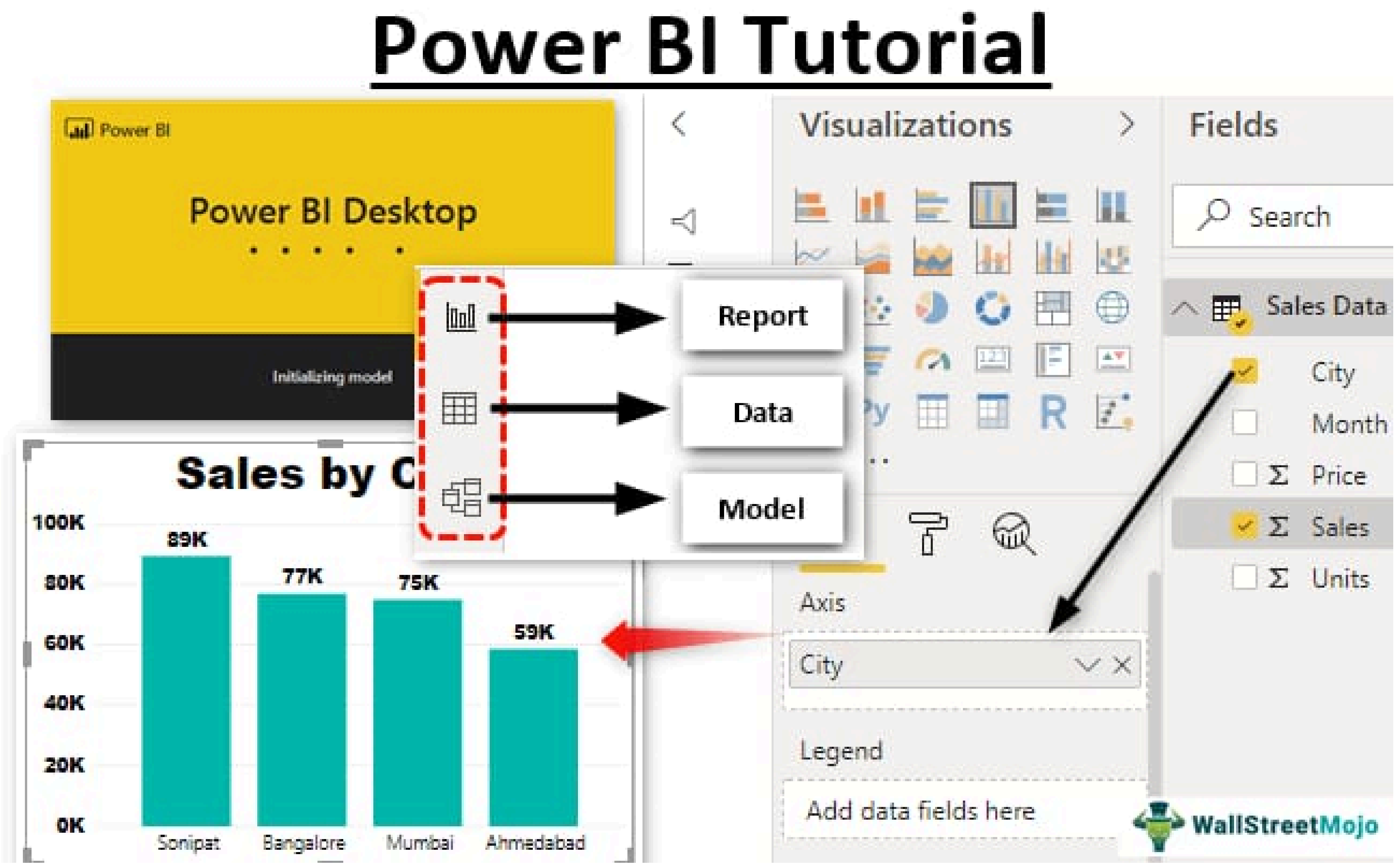
* Pie Chart

* Line Chart

* Table Visualization

Step 6: Design Report Dashboard





You can:

* Resize visuals

* Change colors

* Add titles

* Arrange charts

This creates a *Power BI Dashboard*.

7. Sample Output

A Power BI report showing:

* Sales by Region (Bar Chart)

* Profit Distribution (Pie Chart)

* Monthly Sales Trend (Line Chart)

8. Result

Successfully learned how to use *Power BI Desktop*, import data, and create basic visualizations.

9. Viva Questions

1. What is Power BI?

2. What is the file extension of Power BI reports?

3. What is Power Query Editor?

4. What are dashboards in Power BI?

5. What are visualizations?

Experiment2:Preparing Data in Power BI Design

Aim

To import, clean, and transform data using *Power Query Editor* in *Microsoft Power BI Desktop*.

Software Required

* Microsoft Power BI Desktop

* Excel Dataset

Theory

Power BI is a *Business Intelligence tool* used to analyze and visualize data. Before creating reports, the data must be cleaned and transformed.

This process is done using *Power Query Editor*, which allows users to:

* Remove unwanted columns

* Change data types

* Filter rows

* Split columns

* Prepare data for visualization

Procedure

Step 1: Open Power BI Desktop*

Open *Microsoft Power BI Desktop*.

You will see the *report design window*.

Step 2: Click Get Data

1. Go to *Home Tab*

2. Click *Get Data*

3. Choose *Excel Workbook*

The *Get Data* button is used to connect Power BI to different data sources like Excel, SQL Server, or Web. ([Graphed][1])

Step 3: Import Dataset*

1. Select the Excel file

2. Click *Open*

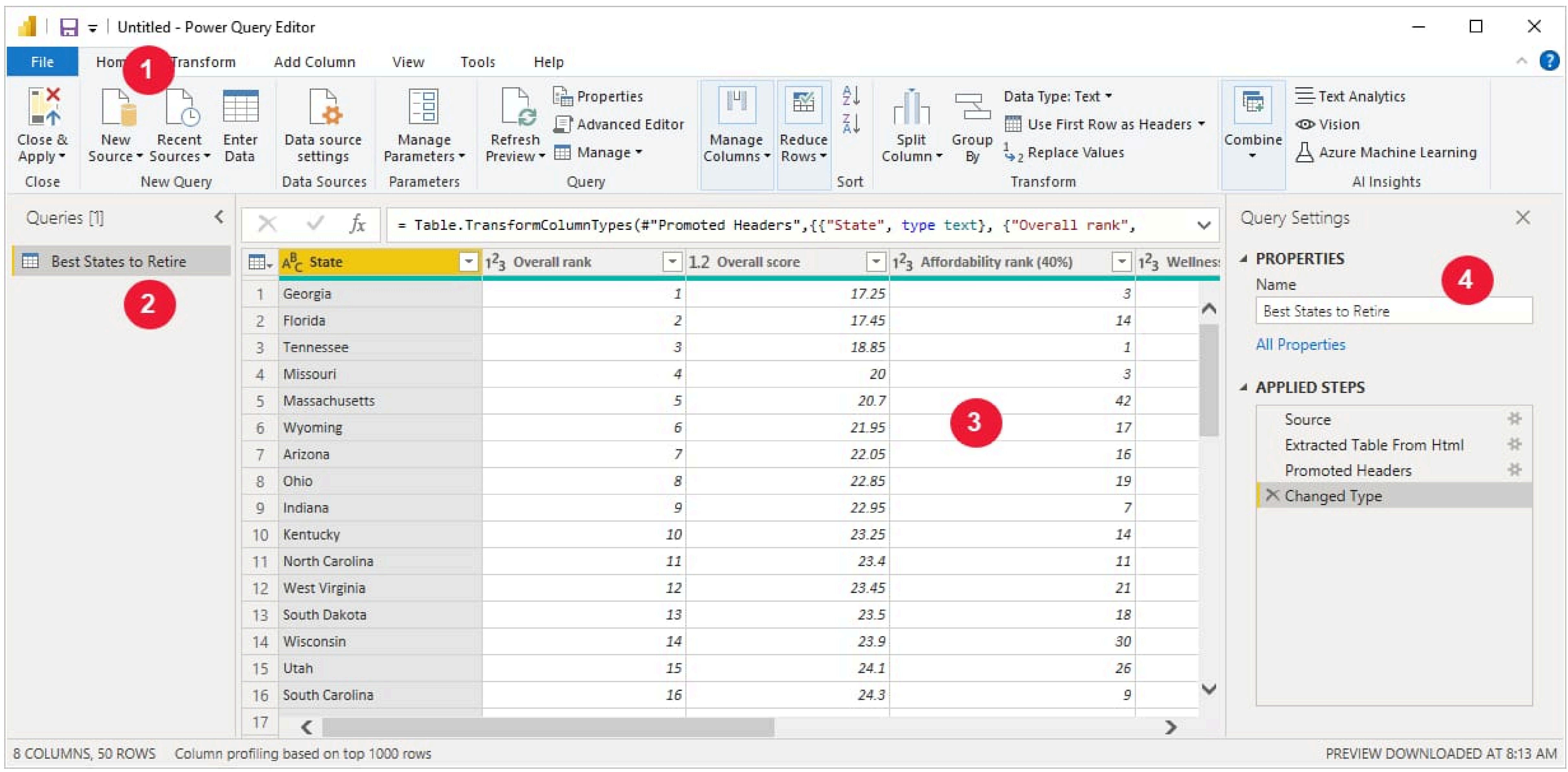
3. The *Navigator Window* will appear

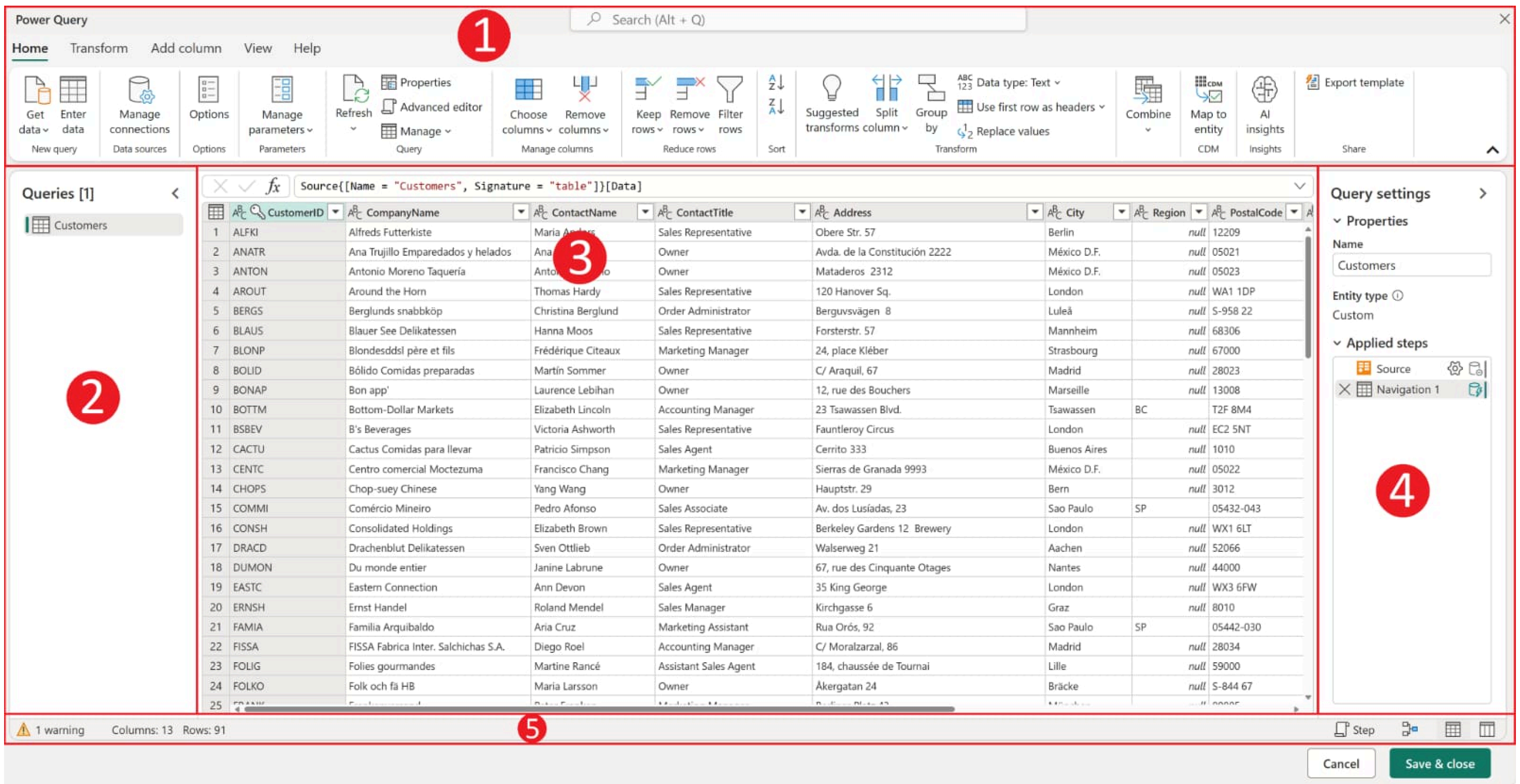
Step 4: Open Power Query Editor

1. Select the table

2. Click *Transform Data*

This opens the *Power Query Editor* where data preparation is done.





Step 5: Remove Unwanted Columns*

1. Select the column

2. Right click

3. Click *Remove*

Step 6: Filter Data

1. Click the *filter icon* in the column header

2. Select the required values

Step 7: Load Data

1. Click *Close & Apply*

2. The prepared data loads into the report page.

Result

The dataset was successfully *imported, cleaned, and prepared using Power Query Editor in Microsoft Power BI Desktop*.

Viva Questions

1. What is Power BI?

2. What is Power Query Editor?

3. What is data transformation?

4. What is the use of Get Data?

5. What is Close & Apply in Power BI?

Experiment 3:LOADING DATA IN POWER BI DESKTOP

1. Aim

To learn how to *load and import data into Power BI Desktop using the Get Data option*.

2. Objectives

After completing this experiment, students will be able to:

* Open *Power BI Desktop*

* Import datasets from *Excel / CSV*

* Use *Navigator window*

* Load data into Power BI for visualization

Power BI allows users to connect to different sources such as Excel, CSV, databases, and web data to analyze information and create reports. ([edhohlbein.com][1])

3. Software Required

* Microsoft Power BI Desktop

* Windows Computer

* Sample dataset (Excel file)

4. Procedure

Step 1: Open Power BI Desktop

Screen Image

```
---------------------------------
|      Power BI Desktop      |
|                            |
| Get Data    Recent Files   |
|                            |
|     Open Other Reports     |
---------------------------------
```

1. Click *Start Menu*

2. Search *Power BI Desktop*

3. Open the application

Step 2: Click Get Data

Screen Image

Home Tab

| File | Home | Insert | Modeling |

[Get Data]

1. Go to *Home tab*

2. Click *Get Data*

3. List of data sources will appear

Step 3: Select Data Source

Screen Image

Get Data Window

Common Data Sources

Excel Workbook

CSV File

SQL Server

Web

More...

----

1. Select *Excel Workbook*

2. Click *Connect*

Step 4: Select Dataset File

Screen Image

File Explorer

Documents

Downloads

Desktop

File: SalesData.xlsx

[Open]

1. Browse your computer

2. Select dataset file

3. Click *Open*

Step 5: Navigator Window

Screen Image

Navigator

Tables Available

☑ Sheet1

☐ Sheet2

Preview of Data Table

[Load]   [Transform Data]

1. Navigator window appears

2. Select *Sheet1 or table*

3. Preview the data

Step 6: Load Data

1. Click *Load*

2. Power BI will import the dataset

3. Data appears in *Fields panel*

Step 7: Data View

Screen Image

Power BI Interface

Fields

Sales

Product

Date

Profit

Report Canvas

----

1. Loaded data will appear on the right side Fields panel

2. Now you can create charts, tables, and dashboards

5. Output

The dataset is successfully loaded into Power BI Desktop and ready for report creation.

6. Result

Thus, the experiment " Loading Data in Power BI" was successfully completed.

Viva Questions

1. What is Power BI?
2. What is Get Data in Power BI?
3. What is the Navigator window?
4. Name any three data sources used in Power BI.
5. What is the difference between Load and Transform Data?

Experiment 4:DATA MODELING IN POWER BI DESKTOP

Aim

To understand and perform *data modeling in Power BI Desktop by creating relationships between tables using Model View*.

Requirements

* Computer with *Microsoft Power BI Desktop installed*

* Sample dataset (Excel/CSV)

* Basic knowledge of Power BI

Theory

Data modeling in Power BI means *creating relationships between different tables to analyze data effectively*.

The *Model View* shows:

* Tables

* Columns

* Relationships between tables

Relationships allow Power BI to combine data from multiple tables and create meaningful reports.

Procedure

Step 1: Open Power BI Desktop

1. Open *Power BI Desktop*.

2. Click *Get Data*.

3. Select *Excel / CSV / Database*.

4. Load the dataset.

Step 2: View the Loaded Tables

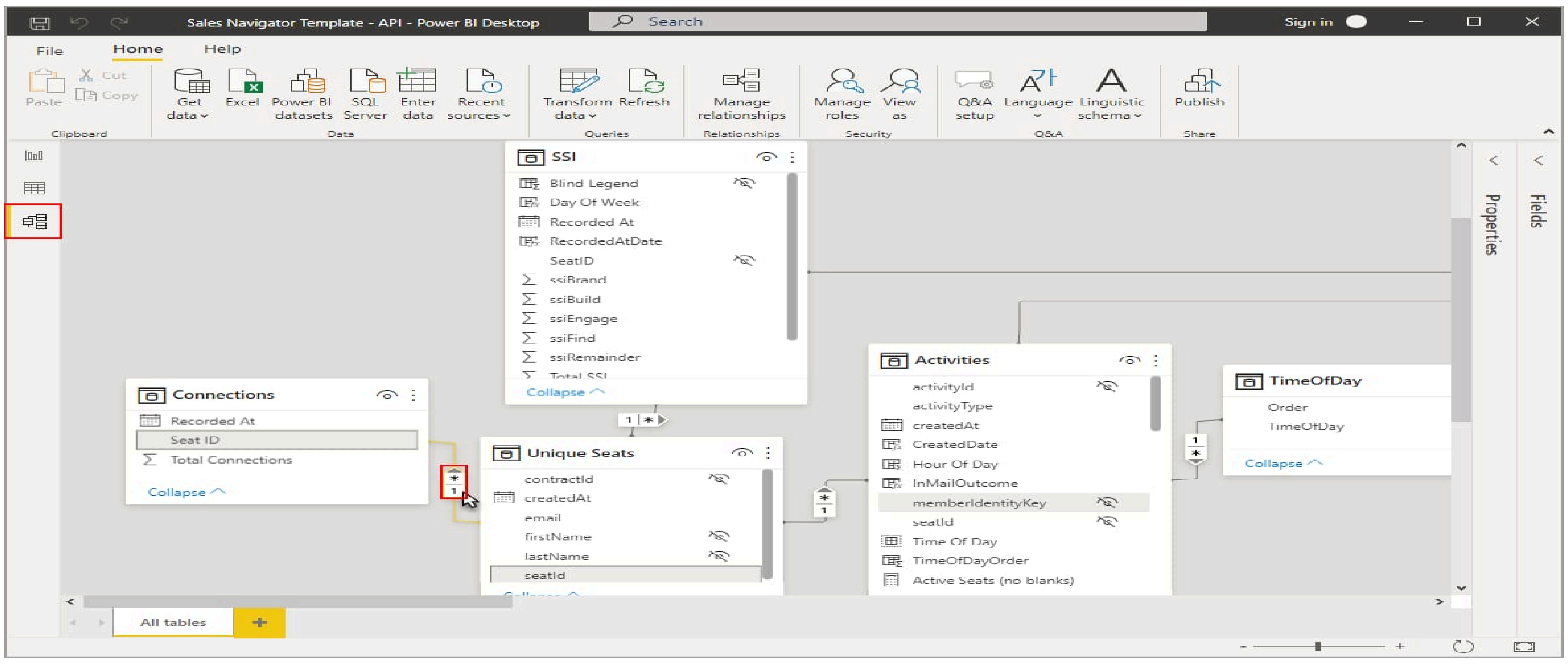
1. After loading data, tables appear in the *Fields Pane*.

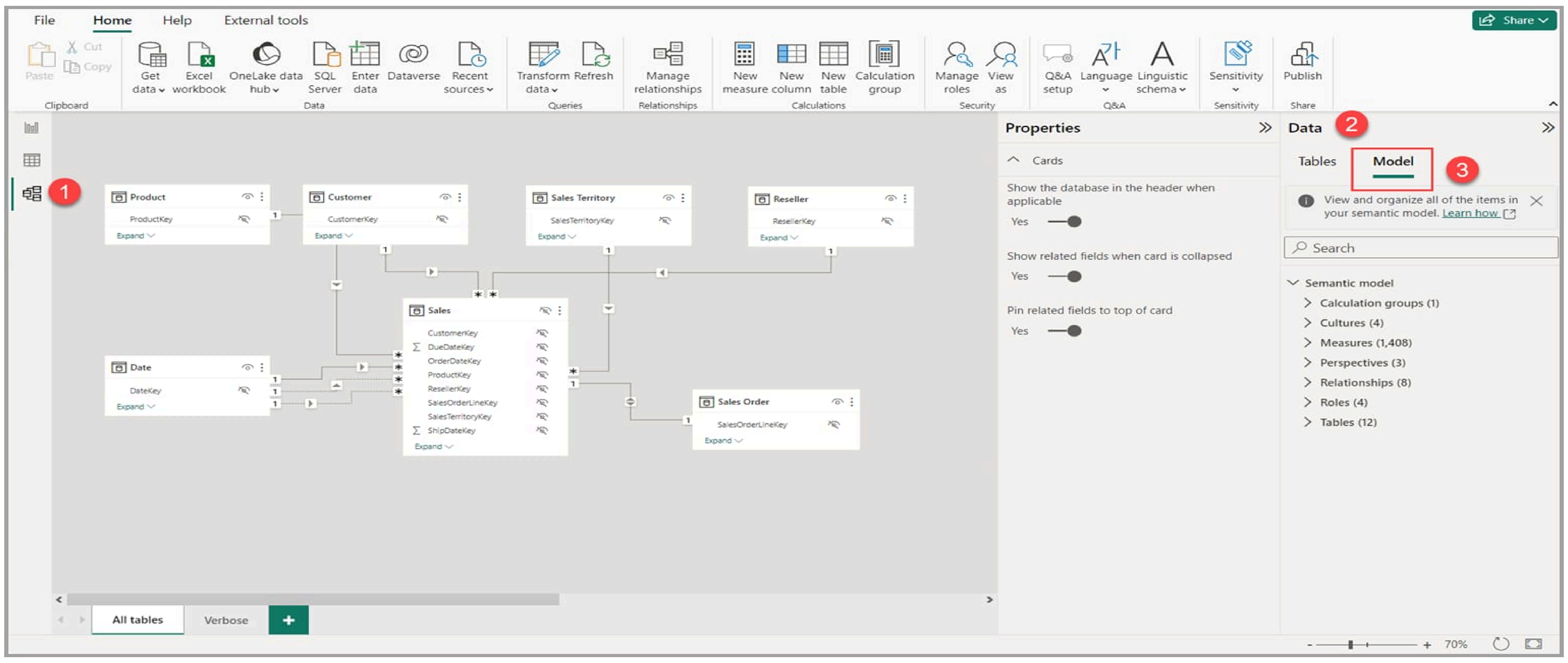
2. Go to *Data View* to see the records.

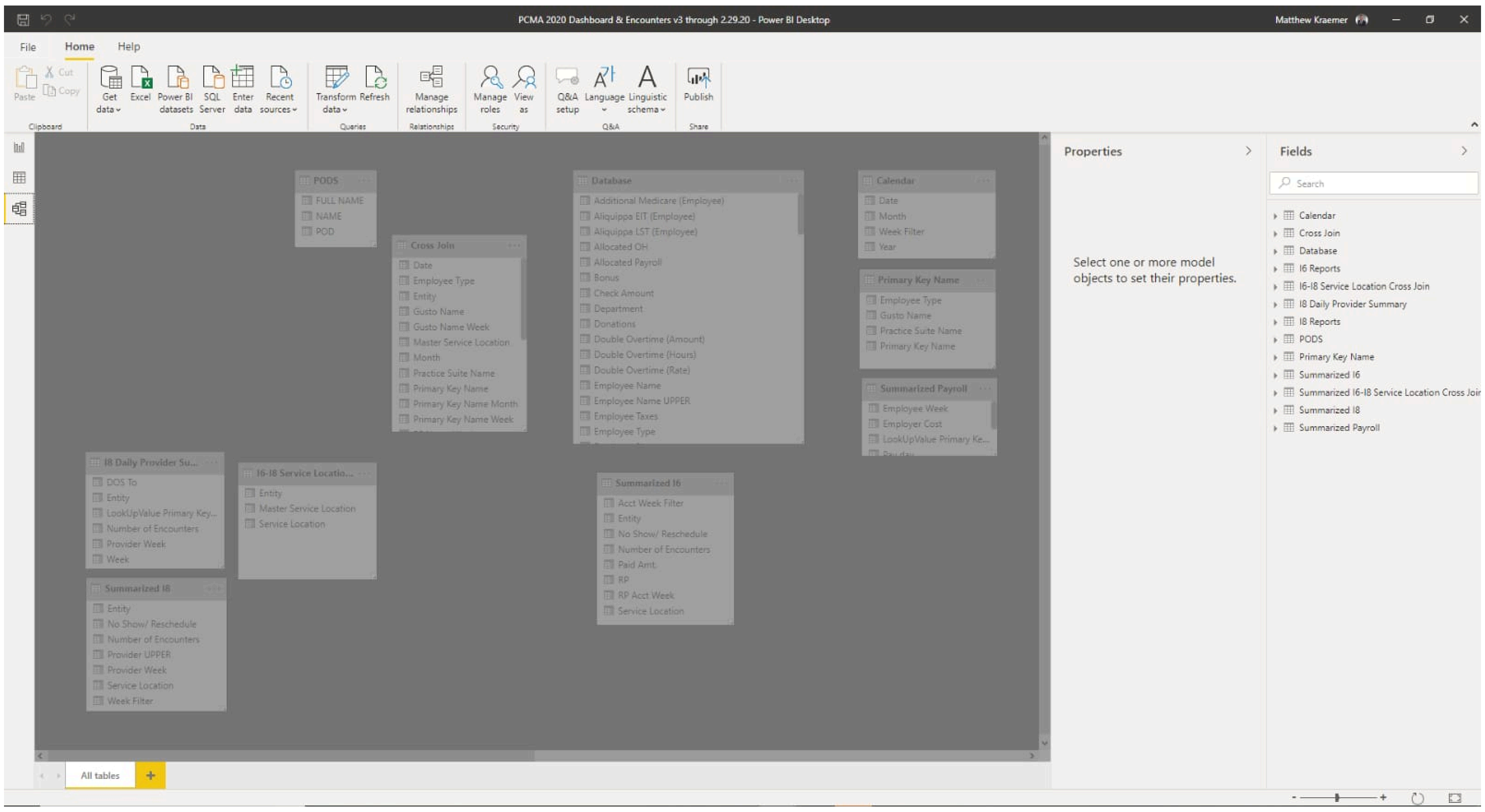
Step 3: Open Model View

1. On the *left side panel, click the **Model icon*.

2. The Model View displays all tables and their relationships.







In this view you can see:

* Tables

* Columns

* Relationship lines connecting tables.

Power BI uses these relationships to filter and analyze data across tables.

Step 4: Create a Relationship

1. In *Model View*, select a column from one table.

2. *Drag the column* to the matching column in another table.

3. A *relationship line* will appear between the tables.

Example:

* CustomerID → Customer Table

* CustomerID → Sales Table

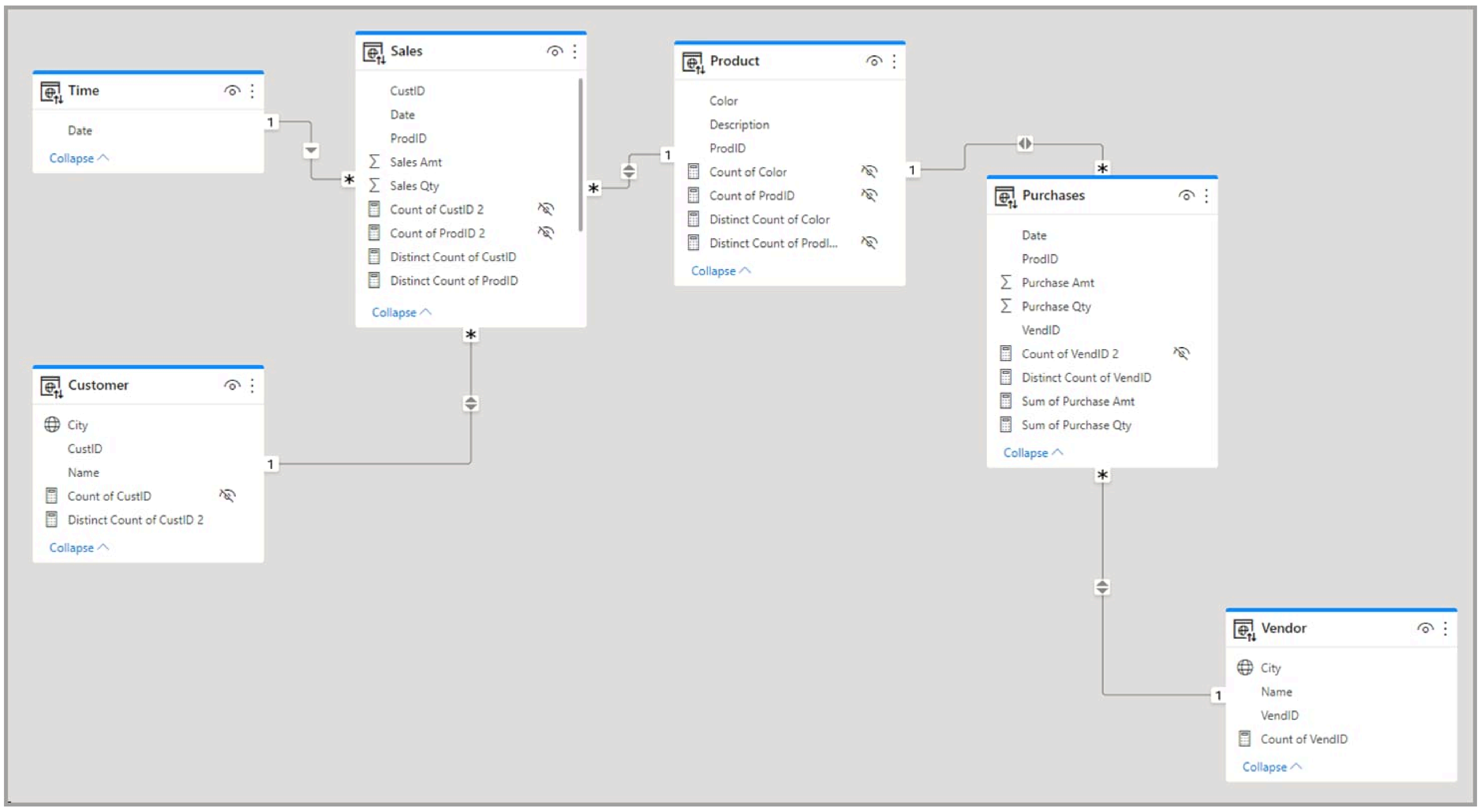
Step 5: Manage Relationships

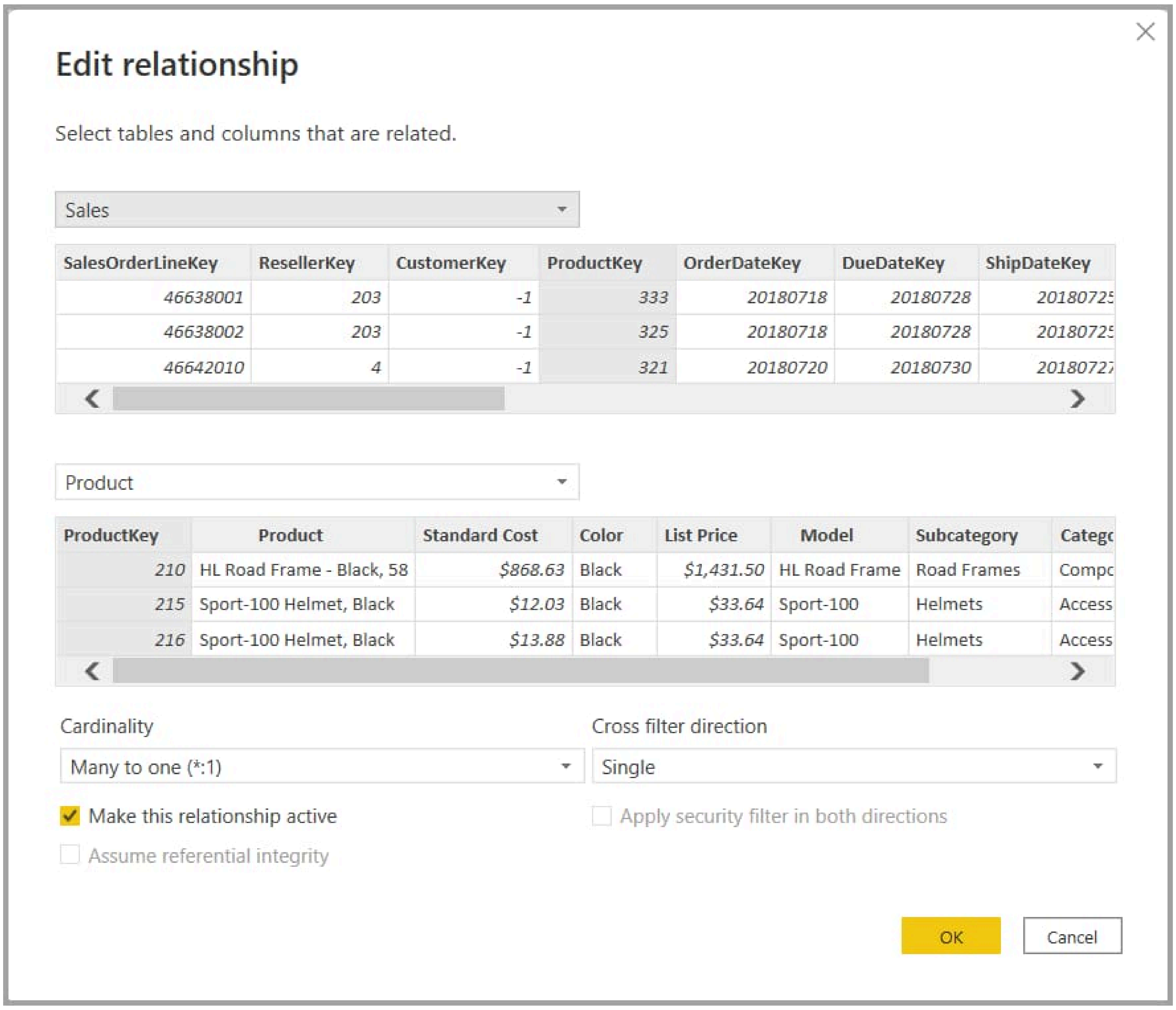
1. Go to *Modeling Tab*.

2. Click *Manage Relationships*.

3. Click *New* to create a relationship.

Step 6: Edit Relationship





In the *Edit Relationship window*, configure:

* Table

* Column

* Cardinality

* Cross filter direction

Then click *OK*.

Types of Relationships

Relationship Type	Description
One to One	Each row in one table relates to one row in another
One to Many	One record relates to many records
Many to Many	Many records relate to many records

Result

Successfully created *data model relationships between tables in Power BI Desktop using Model View

EXPERIMENT 5:ADVANCED DATA MODELING IN POWER BI DESKTOP

1. Aim

To learn **advanced data modeling in Power BI Desktop**, including creating relationships between tables and implementing a **Star Schema data model**.

2. Video Reference

Use this tutorial to follow the lab steps.

3. Software Requirements

- Microsoft Power BI Desktop
- Windows Operating System
- Sample dataset (Excel/CSV)

4. Theory

Data Modeling

Data modeling organizes data into tables and defines relationships between them so that analysis and visualization can be performed efficiently. A common approach in Power BI is the **Star Schema**, where one fact table connects to multiple dimension tables.

Types of Tables

Fact Table

- Contains numerical or measurable data
- Example: Sales, Revenue, Quantity

Dimension Table

- Contains descriptive information
- Example: Product, Customer, Date, Region

5. Procedure

Step 1: Open Power BI Desktop

1. Launch **Power BI Desktop**
2. Click **Get Data**

3. Choose the data source (Excel/CSV)

Power BI Get Data Window

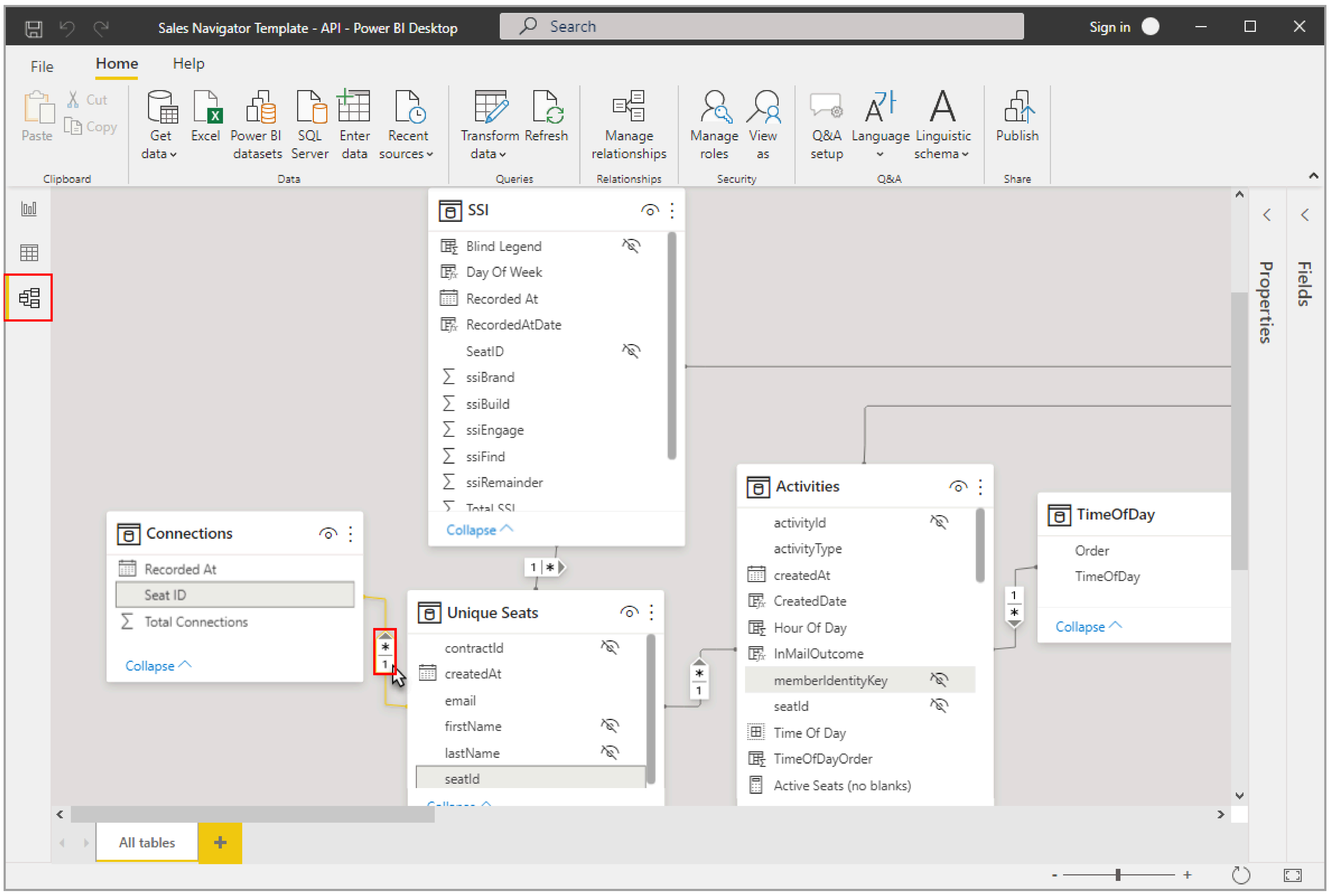
Step 2: Load Dataset

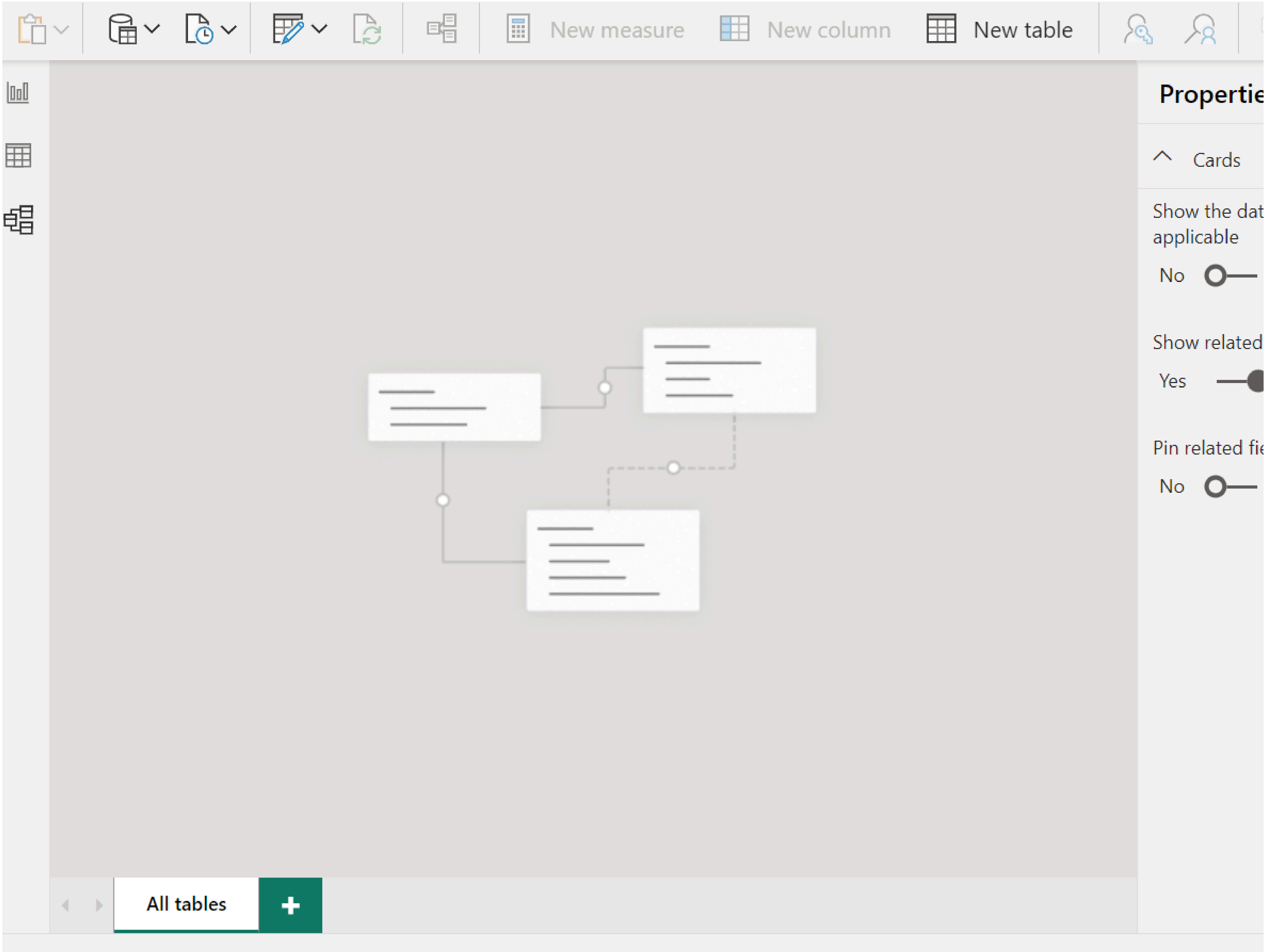
1. Select the dataset file
2. Click **Load**
3. Power BI imports the tables into the workspace.

Step 3: Open Model View

1. Click **Model View** on the left panel.
2. This view shows tables and relationships between them.

Model View in Power BI





Model view helps users visualize relationships between tables in a semantic data model.

Step 4: Create Relationships

1. Drag a column from one table.
2. Drop it onto the related column in another table.

Example relationships:

CustomerID → Customer Table

ProductID → Product Table

DateID → Date Table

Table Relationship Example

In Power BI, relationships are typically **one-to-many**, where a dimension table connects to multiple records in a fact table.

Step 5: Build Star Schema Model

Arrange the tables so that:

- Fact table is in the center
- Dimension tables surround the fact table

Example tables:

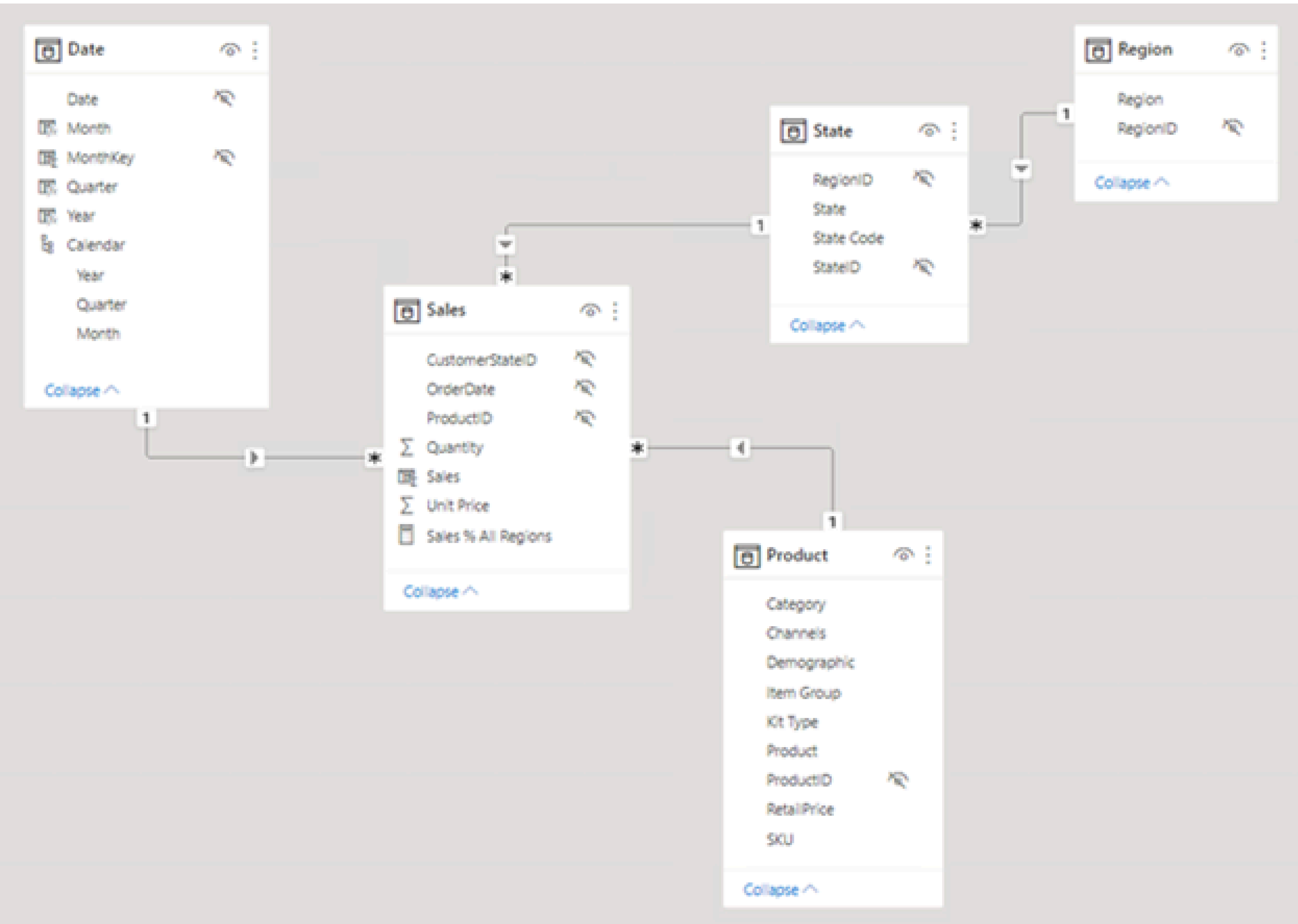
Fact Table

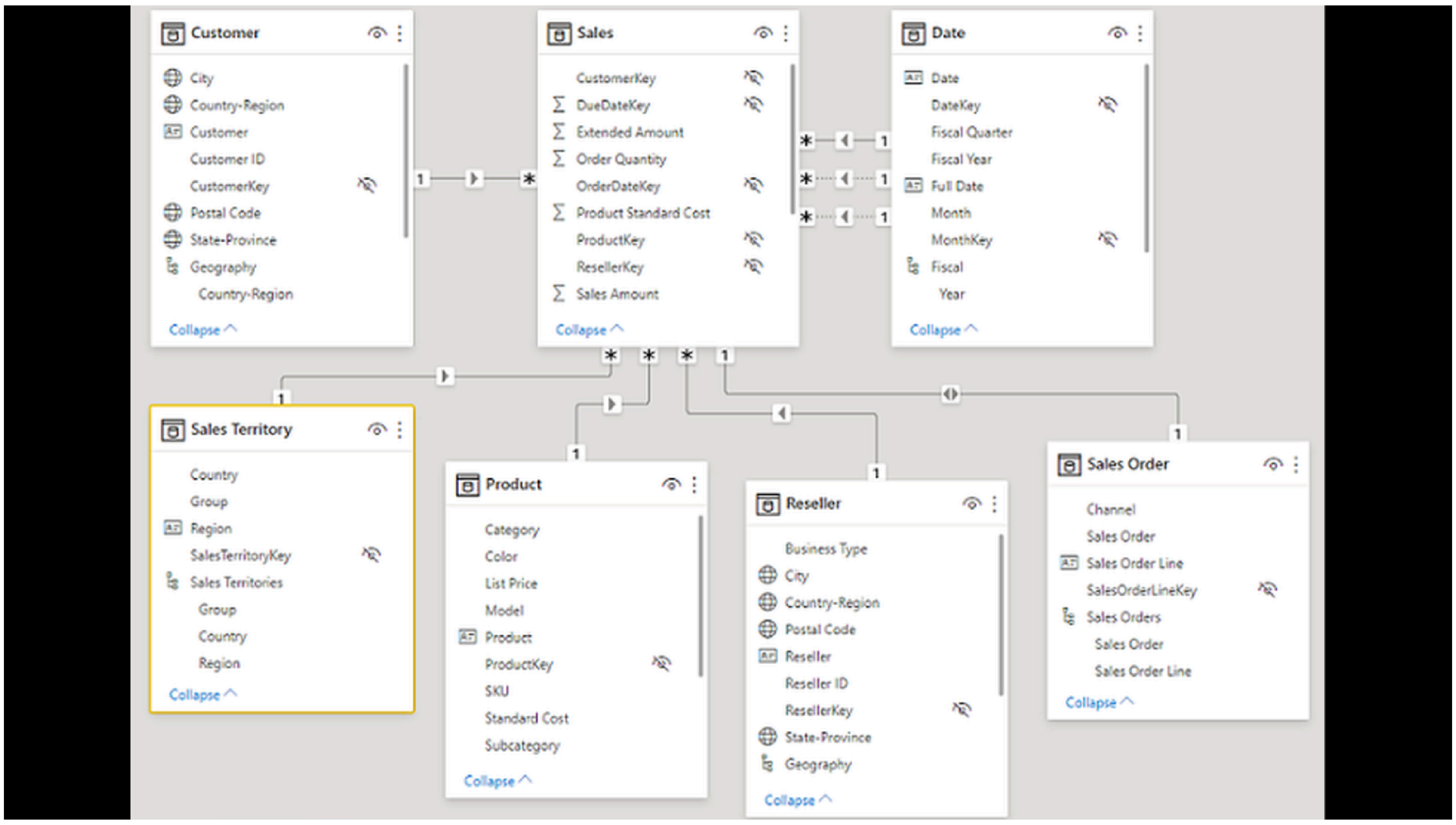
- Sales

Dimension Tables

- Product
- Customer
- Date
- Region

Star Schema Data Model





In a star schema, dimension tables filter the fact table using one-to-many relationships.

6. Result

Successfully created an **advanced data model in Power BI Desktop** by establishing table relationships and implementing a star schema structure.

7. Conclusion

Advanced data modeling improves the efficiency of reports and dashboards in Power BI. By organizing tables into fact and dimension tables, users can create scalable and high-performance analytical models.

EXPERIMENT 6: USING DAX in Power BI DESTOP

1. AIM

DAX (Data Analysis Expressions) is a formula language used in Power BI for creating custom calculations in tables and columns. It consists of functions, operators, and values to perform advanced data analysis.

2. Objective

- Learn how to import data from Excel into Power BI.
- Understand the difference between **Measures** and **Calculated Columns**.
- Implement basic DAX functions: COUNT, SUM, DISTINCTCOUNT, and IF.
- Learn how to use **Quick Measures** for faster calculations.

3. Data Setup & Import

1. Open Power BI Desktop.
2. Go to **Home > Excel workbook**.
3. Select the Orders and Customers datasets.
4. In the Navigator window, check the boxes for the sheets and click **Load**.
5. Use the **Table View** on the left sidebar to inspect your data.

4. Experiment 1: Creating a Basic Measure (COUNT)

Goal: Calculate the total number of orders.

1. Right-click the Orders table in the Data pane.
2. Select **New measure**.
3. Enter the following DAX formula in the formula bar:

   Code snippet

```
Total Orders = COUNT(Orders[Order ID])
```

4. **Explanation:** The COUNT function counts the number of cells in a column that contain non-blank values

5. Experiment 2: Using SUM Function

Goal: Calculate the total quantity of items ordered.

1. Create a **New measure** in the Orders table.
2. Enter the formula:

Code snippet

```
Total Quantity = SUM(Orders[Quantity])
```

3. **Note:** To format the result to zero decimal places, go to the **Measure tools** tab and change the decimal places from 'Auto' to 0

6. Experiment 3: Mathematical Operations (Total Profit)

Goal: Calculate profit by subtracting cost from sales and multiplying by quantity.

1. Create a **New measure**:

   Code snippet

```
Total Profit = SUMX(Orders, (Orders[Selling Price] - Orders[Product Cost]) * Orders[Quantity])
```

7. Experiment 4: Calculated Columns vs. Measures

Feature	Measure (Dynamic)	Calculated Column (Static)
Calculation	Performed at the time of visualization (CPU intensive).	Performed during data load (Memory intensive).
Row Context	Does not see individual rows unless using 'X' functions.	Operates on a row-by-row basis .
Visibility	Only visible in reports/charts.	Visible in the Data/Table view.

Task: Create a "Profit Column" in Table View:

1. Go to **Table View**.
2. Click **New Column** and enter:

Code snippet

```
Profit Column = (Orders[Selling Price] - Orders[Product Cost]) * Orders[Quantity]
```

8. Experiment 5: Conditional Logic (IF Function)

Goal: Categorize orders as "Small" or "Big" based on quantity.

1. Create a **New column** in the Orders table.
2. Enter the formula:

Code snippet

```
Order Category = IF(Orders[Quantity] < 30, "Small Order", "Big Order")
```

3. **Explanation:** The IF function checks a condition. If true, it returns the first result; if false, it returns the second.

9. Conclusion

By completing these exercises, you have learned to perform row-level calculations using Columns and aggregate-level calculations using Measures. DAX allows for deep insights, such as identifying which weekdays have the highest sales or which products are most profitable .

Experiment7: Using Complex DAX Queries in Power BI Desktop

Aim

To understand and implement complex DAX (Data Analysis Expressions) queries for advanced data analysis in Power BI Desktop.

Requirements

Power BI Desktop

Sample dataset (Sales / Financial data)

Basic knowledge of Power BI

Theory

DAX (Data Analysis Expressions) is a formula language used in Power BI to create calculations, measures, and custom tables. ?

Microsoft Learn

It includes functions, operators, and expressions used to analyze data. ?

Microsoft Learn

DAX enables complex calculations like time intelligence, filtering, and dynamic aggregations. ?

DataCamp

Key Concepts of Complex DAX

Measures – Dynamic calculations (e.g., total sales)

Calculated Columns – Row-level calculations

Filter Context – Filters applied during calculation

Row Context – Current row processing

👉Complex DAX mainly uses:

CALCULATE()

FILTER()

SUMX(), AVERAGEX()

Time intelligence functions

Procedure

Step 1: Load Dataset

Open Power BI Desktop

Click Get Data → Excel/CSV

Load dataset

Step 2: Open Data Model

Go to Data View / Model View

Check relationships between tables

Step 3: Create Basic Measure

DAX

```
Total Sales = SUM(Sales[SalesAmount])
```

👉This calculates total sales using aggregation.

Step 4: Create Complex DAX Measure

Example 1: Sales in Previous Quarter

DAX

```
Previous Quarter Sales =
CALCULATE(
    SUM(Sales[SalesAmount]),
    PREVIOUSQUARTER(Calendar[Date])
)
```

👉Uses time intelligence + filter context

Example 2: Filtered Sales

DAX

```
Filtered Sales =
CALCULATE(
    [Total Sales],
```

```
    FILTER(Sales, Sales[Region] = "East")
)
```

👉Applies custom filter using FILTER()

Step 5: Use Iterator Function

```DAX
Average Sales =
AVERAGEX(
    Sales,
    Sales[SalesAmount]
)
```

👉Iterates row by row

Step 6: Create Conditional Logic

```DAX
Profit Category =
IF(
    [Total Sales] > 100000,
    "High",
    "Low"
)
```

Step 7: Create DAX Query (Advanced)

Basic structure:

```DAX
EVALUATE Sales
```

👉Returns full table

Advanced query:

DAX

```
EVALUATE
FILTER(
   Sales,
   Sales[SalesAmount] > 5000
)
```

👉Returns filtered table data ?

daxstudio.org

Step 8: Use CALCULATE with Multiple Filters

DAX

```
Complex Measure =
CALCULATE(
   SUM(Sales[SalesAmount]),
   Sales[Region] = "West",
   Sales[Category] = "Electronics"
)
```

Step 9: Add Measures to Report

Drag created measures into visuals

Use charts, tables, KPI cards

Step 10: Analyze Output

Apply slicers (date, region)

Observe dynamic changes

Sample Output

Your report should include:

Total Sales

Filtered Sales

Previous Quarter Sales

KPI indicators

Interactive visuals

Result

Successfully created and used complex DAX queries to perform advanced data analysis in Power BI Desktop.

Viva Questions

What is DAX?

Difference between measure and calculated column?

What is CALCULATE function?

What is filter context?

What are iterator functions?

Conclusion

Complex DAX queries allow users to perform advanced analytics and dynamic calculations in Power BI. Understanding context, filters, and functions is essential for mastering DAX and building powerful reports.

Experiment8: Designing a Report in Power BI Desktop

Aim

To design an interactive and visually appealing report using Power BI Desktop.

Requirements

Power BI Desktop software

Dataset (Excel / CSV file)

Basic knowledge of data visualization

Theory

Power BI is a business analytics tool used to create reports and dashboards. It allows users to transform raw data into meaningful insights using visuals like charts, tables, and maps.

A good report design focuses on:

Clear layout

Proper use of visuals

Easy navigation

Interactive filters

Well-designed reports help users quickly understand key insights and make decisions efficiently. ?

Power BI Consulting

Procedure

Step 1: Open Power BI Desktop

Launch Power BI Desktop

Click on Get Data

Step 2: Import Dataset

Select Excel/CSV file

Click Load

👉Power BI supports multiple data sources and loads them into the model for analysis. ?

almabetter.com

Step 3: Data Transformation

Click Transform Data

Clean the dataset:

Remove null values

Rename columns

Change data types

Step 4: Create Report Layout

Go to Report View

Plan layout:

Title at top

Charts in middle

Filters on side

👉Good layout helps users scan report within seconds. ?

Power BI Consulting

Step 5: Add Visualizations

1. Add Title

Insert → Text Box

Type report title

2. Add Charts

Select visualization type:

Bar chart

Line chart

Pie chart

Drag fields into values

3. Add Table / Matrix

Display detailed data

👉Visuals are created by dragging fields onto the report canvas. ?

almabetter.com

Step 6: Apply Filters and Slicers

Add slicer from visualization pane

Use fields like:

Date

Category

Region

👉Filters help users interact with data dynamically. ?

Microsoft Learn

Step 7: Format Report

Change theme (View → Themes)

Adjust:

Colors

Fonts

Background

👉Themes ensure consistent design across the report. ?

ExcelDemy

Step 8: Arrange Visuals

Align charts properly

Maintain spacing

Highlight key KPIs

Step 9: Save Report

File → Save (.pbix format)

Sample Output (What Your Report Should Include)

Title (e.g., Sales Report)

Charts (Bar, Line, Pie)

KPI Cards

Filters (Slicers)

Clean layout

Result

Successfully designed an interactive report in Power BI Desktop with proper layout, visuals, and filters.

Viva Questions

What is Power BI?

What are slicers in Power BI?

Difference between report and dashboard?

What is a visualization?

Why is report design important?

Conclusion

Power BI enables users to design professional reports by combining data, visuals, and interactivity. Proper design improves readability and helps users make better decisions.

Experiment9: Data Analysis in Power BI Desktop

Aim

To perform data analysis using Power BI Desktop by importing, transforming, modeling, and visualizing data.

Requirements

Power BI Desktop

Dataset (Excel / CSV / Database)

Basic knowledge of data handling

Theory

Power BI Desktop is a business intelligence tool used to analyze and visualize data. It allows users to connect to multiple data sources, clean and transform data, and create interactive reports.

Microsoft Learn

Data Analysis Workflow in Power BI

Data Import

Data Cleaning (Power Query)

Data Modeling

Data Analysis (DAX)

Data Visualization

👉This step-by-step workflow helps transform raw data into meaningful insights. ?

DataCamp

Procedure

Step 1: Open Power BI Desktop

Launch Power BI Desktop

Click on Get Data

Step 2: Import Data

Select source (Excel / CSV / SQL)

Click Load

👉Power BI supports multiple data sources for analysis. ?

Power BI Consulting

Step 3: Data Cleaning (Power Query)

Click Transform Data

Perform:

Remove null values

Change data types

Filter unwanted data

Rename columns

👉Cleaning ensures accurate analysis results. ?

Microsoft Learn

Step 4: Data Modeling

Go to Model View

Create relationships between tables

Define primary and foreign keys

👉Proper modeling improves analysis efficiency. ?

GeeksforGeeks

Step 5: Data Analysis using DAX

Example Measure:

DAX

Total Sales = SUM(Sales[SalesAmount])

Example Profit Margin:

DAX

Profit Margin = SUM(Sales[Profit]) / SUM(Sales[Sales])

👉DAX is used to create calculated columns, measures, and tables for analysis. ?

DataCamp

Step 6: Create Visualizations

Go to Report View

Add visuals:

Bar chart

Line chart

Pie chart

Table / Matrix

👉Visualizations help identify trends and patterns in data. ?

Microsoft Learn

Step 7: Apply Filters and Slicers

Add slicers (Date, Region, Category)

Enable interactive filtering

👉Filters allow dynamic data exploration. ?

GeeksforGeeks

Step 8: Data Analysis Insights

Identify:

Highest sales region

Monthly trends

Profit vs Loss

Use visuals to derive insights

Step 9: Format Report

Apply themes

Adjust colors and layout

Add titles and labels

Step 10: Save and Publish

Save file (.pbix)

Publish to Power BI Service

Sample Output

Your report should include:

KPI Cards (Total Sales, Profit)

Charts (Sales by Region, Monthly Trend)

Slicers (Date, Category)

Clean dashboard layout

Result

Successfully performed data analysis in Power BI Desktop by transforming raw data into meaningful insights using DAX and visualizations.

Viva Questions

What is Power BI?

What is Power Query?

What is DAX?

What is data modeling?

What are slicers?

Conclusion

Power BI Desktop provides a complete platform for data analysis, including data preparation, modeling, and visualization. It enables users to convert raw data into actionable insights for better decision-making.

Experiment10. Publishing and Sharing Power BI Content

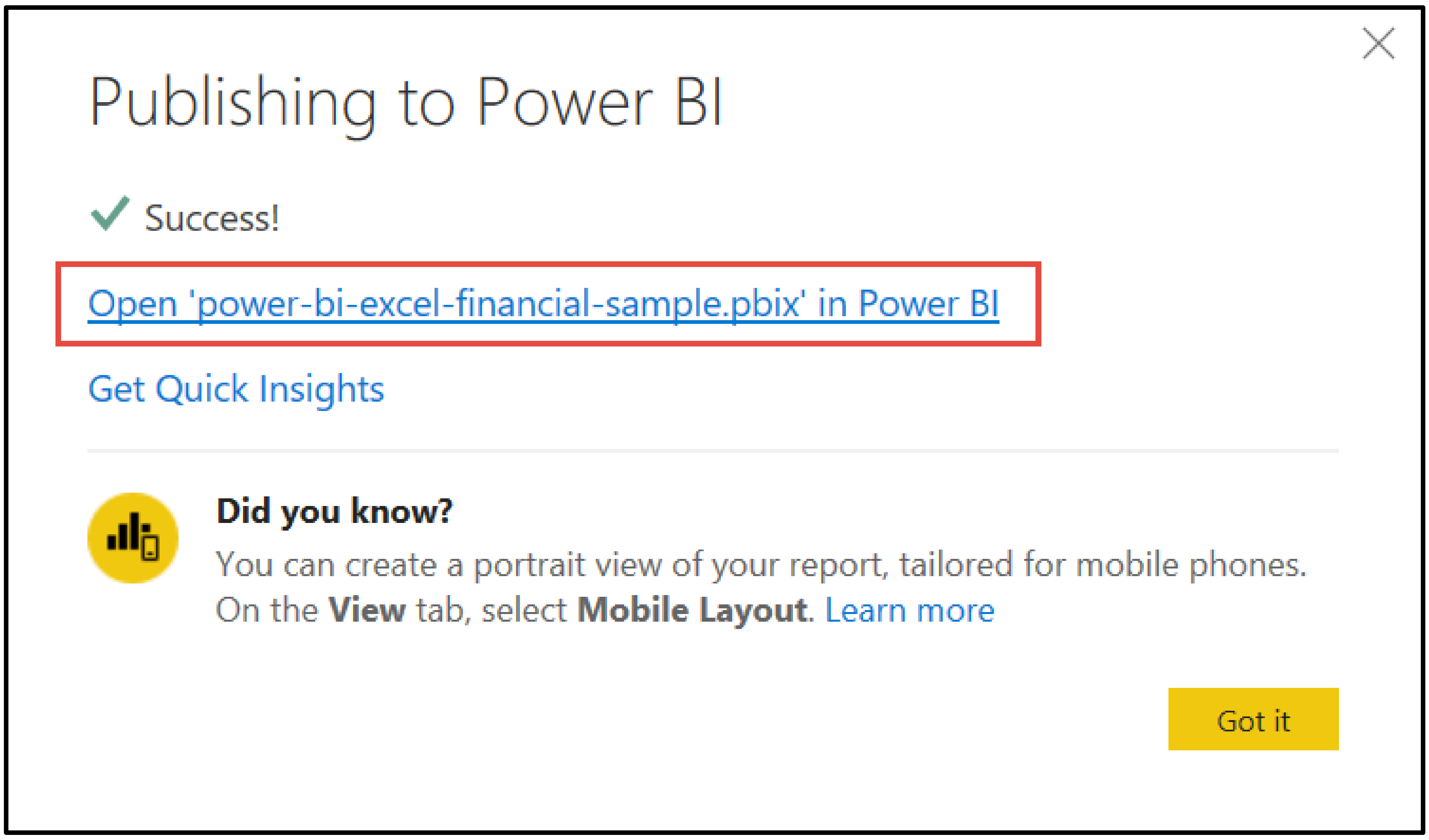
Publish to the Power BI service to share

To share your report with your manager and colleagues, publish it to the Power BI service. When you share with colleagues that have a Power BI account, they can interact with your report, but can’ t save changes.

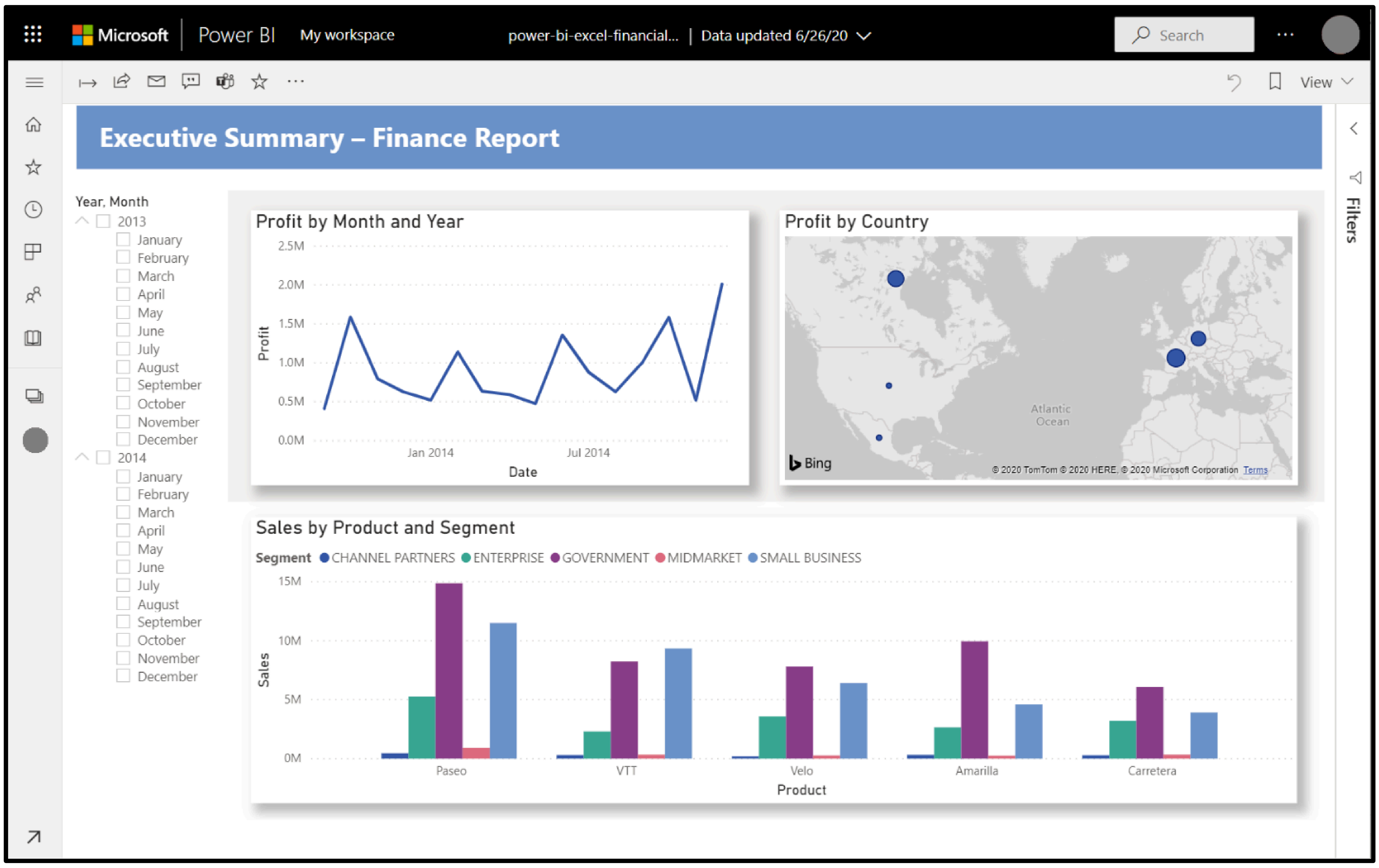
1. In Power BI Desktop, select **Publish** on the **Home** ribbon.

You may need to sign in to the Power BI service. If you don't have an account yet, you can sign up for a free trial.

2. Select a destination such as **My workspace** in the Power BI service > **Select**.
3. Select **Open 'your-file-name' in Power BI**.



Your completed report opens in the browser.



4. Select **Share** at the top of the report to share your report with others.